

The missing factor in Conjecture 00000003454

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October 9, 2026

Abstract

For the smooth polynomial $f(x) = (x - 1/2)^2$, the Bernstein approximation at $x = 1/2$ has error exactly $1/(4n)$. Thus the error multiplied by n converges to $1/4$, whereas $f''(1/2) = 2$. This disproves the source's identification of the deviation limit with the second derivative. The unscaled error also has a different limit, namely zero.

1 Statement and interpretation

The source conjecture describes the Voronovskaya asymptotics of Bernstein operators by saying that “the limit of the deviation is the second derivative,” with a rate inverse in n . The standard Bernstein operator is

$$(B_n f)(x) = \sum_{k=0}^n f(k/n) \binom{n}{k} x^k (1-x)^{n-k}, \quad n \geq 1.$$

We examine the usual normalized deviation $n(B_n f - f)$ and, separately, the literal unscaled deviation $B_n f - f$. Neither converges to f'' in our example, even at an interior point and for a polynomial. The issue is the coefficient of the second derivative, not the occurrence of an order $1/n$ approximation error.

2 Exact counterexample

Theorem 1. *For $f(x) = (x - 1/2)^2$ and every positive integer n ,*

$$(B_n f)(1/2) - f(1/2) = \frac{1}{4n}.$$

Consequently, $n((B_n f)(1/2) - f(1/2))$ converges to $1/4$, and the unscaled deviation converges to 0. Both limits differ from $f''(1/2) = 2$.

Proof. The binomial variance identity gives, for $0 \leq x \leq 1$,

$$\sum_{k=0}^n (k/n - x)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n}.$$

At $x = 1/2$, the left side is exactly $(B_n f)(1/2)$, while the right side is $1/(4n)$. Since $f(1/2) = 0$, this proves the displayed formula. Multiplying by n gives the constant $1/4$. Without that multiplication, the error tends to zero. Direct differentiation gives $f'(x) = 2(x - 1/2)$ and $f''(x) = 2$. The asserted limit is therefore false under either reading of the deviation. $\square$

The coefficient matters. The usual Voronovskaya expression contains the factor $x(1-x)/2$ in front of $f''(x)$; at the midpoint this factor is $1/8$ and gives $1/4$ for our quadratic. The proof above is an exact finite-sum calculation and does not assume a general Voronovskaya theorem. It does not dispute the classical formula with its factor restored, or any separately specified renormalization that divides out this factor. Neither such factor nor such a renormalization appears in the source.

3 Formal verification

The Lean project uses Mathlib’s actual `bernsteinApproximation` on continuous functions on $[0, 1]$, its proved binomial variance identity, real derivatives, and filter limits. In particular:

- `quadratic_smooth` proves the polynomial is infinitely differentiable; `quadratic_second_derivative` evaluates the second derivative at every real point.
- `approximation_at_midpoint` identifies the actual Bernstein sum with $1/(4n)$ for every positive integer n .
- `scaled_deviation_limit` and `deviation_limit` prove the normalized and unscaled limits. The sequences are indexed by $n + 1$ so every approximation order is positive.
- `scaled_second_derivative_limit_false` and `raw_second_derivative_limit_false` state the two contradictions using the actual second derivative, not a substituted numerical surrogate.

No auxiliary numerical computation is required. All final theorems use only Lean’s standard logical axioms, with no unproved placeholders, custom axioms, or native evaluation used as a proof.

Reproduction. In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

The project pins Lean 4.19.0 and an exact Mathlib revision. Run `tectonic main.tex` to rebuild the PDF. `SOURCE.md` preserves the bilingual conjecture.

References

- [1] The Last Math Competition, Conjecture 00000003454.
- [2] The Mathlib Community, Lean 4 mathematical library, revision c44e0c8ee63c, module `Analysis.SpecialFunctions.Bernstein`.